

Abir Sarkar

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I am a Ph.D. student in Statistics and Data Science at **Cornell University**, with prior experience in quantitative model validation at **Capital One**. My research focuses on developing statistically principled methods in **time series econometrics**, **active statistical learning**, and **stochastic optimization**.

Education

Cornell University <i>Ph.D. in Statistics and Data Science, Advisor: Prof. Martin T. Wells</i> <i>Committee Member: Prof. Jelena Bradic</i>	2025–2029 <i>Ithaca, NY</i>
Indian Statistical Institute <i>Master of Statistics, specialization in Finance</i>	2021–2023 <i>Kolkata, India</i>
Indian Statistical Institute <i>Bachelor of Statistics</i>	2018–2021 <i>Kolkata, India</i>

Industry Experience & Internships

Capital One <i>Senior Associate, Data Scientist, Model Validation</i>	Jun 2023–Jul 2025 <i>Bangalore, India</i>
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- Validated Commercial and Industrial, Commercial Real Estate, and Structured Finance probability-of-default and loss-given-default models covering more than \$140 billion in portfolio exposure. Led independent validation of a \$7 billion Municipal LGD model and approximately \$20 billion in Healthcare C&I and Healthcare CRE portfolios.
- Conducted conceptual-soundness reviews, benchmarking, sensitivity analysis, independent model replication, prototype testing, implementation testing, and deployment validation. Contributed to retail-fraud modeling, consumer-loss forecasting, deposit modeling, cannibalization models, and CCAR and CECL stress-testing initiatives. Supported internal governance and regulatory reviews, completed two audit examinations with no findings.

JPMorgan Chase & Co. <i>Market Risk Analyst Intern</i>	Summer 2022 <i>Mumbai, India</i>
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Evaluated the performance of trading and risk models used in foreign exchange and commodities businesses within the Corporate Investment Bank. Developed a quantitative framework for rating models as part of ongoing performance monitoring.

Big Data Summer Institute, University of Michigan <i>Research Intern and Team Lead</i>	Summer 2021 <i>Ann Arbor, MI</i>
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Led a team analyzing Michigan Medicine data and developed missing-data imputation pipelines using regression, multiple imputation, regression trees, random forests, and XGBoost. Used the imputed clinical data to construct and evaluate statistical models for predicting hospitalization outcomes.

Publications

- Is There an AI Bubble? Robust Date-Stamping for Periods of Exuberance**
Abir Sarkar and Martin T. Wells, *Accepted in Frontiers of Mathematical Finance*
We developed a stochastic-volatility-robust ADF framework for date-stamping asset-price bubbles under persistent and time-varying volatility, deriving nuisance-parameter-free asymmetric calibration rules for bubble origination and collapse.
- Asymptotic Breakdown Point Analysis for a General Class of Minimum Divergence Estimators**
Subhrajyoty Roy, Abir Sarkar, Abhik Ghosh, and Ayanendranath Basu, *Accepted in Bernoulli*
We investigated the global robustness of minimum S -divergence estimators under general parametric models, developing asymptotic breakdown-point theory and dimension-free robustness guarantees for a broad class of minimum-divergence estimators as robust alternatives to maximum likelihood.

Submitted Articles

- Adapt or Forget: Provable Tradeoffs between Adam and SGD in Nonstationary Optimization**
Sharan Sahu*, Abir Sarkar*, Cameron J. Hogan*, and Martin T. Wells, *Preprint, Submitted to NeurIPS*
- Double Local-to-Unity: Inference under Nearly Nonstationary Volatility**
Abir Sarkar and Martin T. Wells, *Preprint, Submitted to Journal of Econometrics*
- Empirical Bayes Predictive Density Estimation under Covariate Shift in Large Imbalanced Linear Mixed Models**
Abir Sarkar, Gourab Mukherjee, and Keisuke Yano, *Preprint, Submitted to JASA*

Teaching Experience at Cornell University

ECON 2801: Game Theory—For Economics, Politics, Knowledge, and Rationality

Grader, Cornell University

Fall 2025, 2026

Instructor: *Prof. Kaushik Basu*

STSCI 4981: Sports Analytics in Practice

Teaching Assistant and Grader

Spring 2026

Instructor: *Prof. Martin T. Wells*

STSCI 5080: Probability Models and Inference

Teaching Assistant and Grader

Fall 2026

Instructor: *Prof. Kengo Kato*

Conference Presentations

- **Is There an AI Bubble? Robust Date-Stamping for Periods of Exuberance**
 - The 39th New England Statistics Symposium (NESS), *From Data to Discovery: Statistical Insight in the Age of AI*, University of Connecticut, Storrs, CT (2026).
 - Research featured in [Cornell Chronicle](#).
- **Empirical Bayes Predictive Density Estimation under Covariate Shift in Large Imbalanced Linear Mixed Models**
 - Annual Conference of the International Indian Statistical Association, Kochi, India (2024).
 - International Conference on Statistics and Data Science, Seville, Spain (2025).

Awards and Achievements

- **Director's Award**, Capital One, for contributions to the validation of Commercial C&I and Municipal default models.
- **Statistically Significant Award**, Capital One, for exceptional performance across multiple projects under demanding timelines.
- **INSPIRE Scholarship**, Department of Science and Technology, Government of India, 2018–2023.
- Secured **1st rank** in the **Jagadis Bose National Science Talent Search (JBNSTS)**, 2018.
- Ranked **3rd** in both the B.Stat. (2018–2021) and M.Stat. (2021–2023) cohorts at the Indian Statistical Institute.
- Ranked **4th** and **5th** in the **Simon Marais Mathematics Competition** in 2018 and 2019, respectively.

Technical Skills

Programming: Python, R, SQL, C++, $\text{\LaTeX}$, Git

Language: English, Hindi, Bengali